

Multiple Choice

1. **Answer:** A

Options: A) 0.185 nm; B) 0.110 nm; C) 0.125 nm; D) 0.200 nm; E) 0.095 nm

Note: Correct option: A - 0.185 nm

2. **Answer:** A

Options: A) $6.75 \times 10^{-10}m$; B) $8.22 \times 10^{-11}m$; C) $2.15 \times 10^{-12}m$; D) $1.885 \times 10^7 \text{ m s}^{-1}$; E) $1.22 \times 10^{-9}m$

Note: *Correct option* : A - $6.75 \times 10^{-10}m$

2. **Answer:** A

Options: A) $22.6 \times 10^{\text{f}} - 3 \text{ m}^3 \text{ mol}^{-1}$; B) $0.0226 \text{ m}^3 \text{ mol}^{-1}$; C) $2.26 \times 10^{\text{f}} - 6 \text{ m}^3 \text{ mol}^{-1}$; D) $226 \times 10^{\text{f}} - 6 \text{ m}^3 \text{ mol}^{-1}$; E) $2.26 \times 10^{\text{f}} - 2 \text{ m}^3 \text{ mol}^{-1}$

Note: Correct option: A - $22.6 \times 10^{\text{f}} - 3 \text{ m}^3 \text{ mol}^{-1}$

3. **Answer:** E

Options: A) Proteins determine the rate of photosynthesis in plants.; B) Proteins are used for insulation in cold environments; C) Proteins are the main constituent of cell walls in bacteria.; D) Proteins are responsible for coloration of organisms; E) Proteins serve as structural material and biological regulators.

Note: Correct option: E - Proteins serve as structural material and biological regulators.

4. **Answer:** A

Options: A) 0.311; B) 0.5; C) 1.00; D) 0.05; E) 0.10

Note: Correct option: A - 0.311

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'electronegativity'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '7.51 MeV'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** -3.2 L atm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 6.10 ppm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key